

Track-to-Track fusion with cross-covariances from radar and IR/EO sensor

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Abstract—The Track-to-track fusion (T2TF) problem for estimates from radar and infrared/electro-optical (IR/EO) sensor is studied in this work. For such a problem, the heterogeneous estimates from local trackers (LT) are in different state spaces with various dimensions and are related by a nonlinear relationship with no inverse transformation. For the homogeneous T2TF problem, where the common state model is shared by both LTs in the same state space, the cross-covariance between the local estimation errors, which has been known for some time, needs to be considered in the T2TF. However, such a cross-covariance for heterogeneous T2TF was not available in previous works. In the present work, the derivation of the cross-covariance for heterogeneous LTs of different dimension states is provided, yielding a recursion, by taking into account the relationship between the local state model process noises. A linear minimum mean square (LMMSE) estimator is used for the T2TF. With the cross-covariance involved, the fusion will generate the covariance of the fused estimation error which makes the system consistent as shown in the simulation through Monte-Carlo runs.

I. INTRODUCTION

A centralized information processing system without data reduction could provide an optimal solution in data fusion, however, it requires high communication capacity. Thus, it is not feasible or limited due to real life communication constraints in most cases. In a distributed (decentralized) information processing configuration, each sensor has its own information processing system — local tracker (LT). The track to track fusion (T2TF) is carried out at the fusion center (FC) combining the corresponding estimates from LTs assuming correct track to track association. A homogeneous T2TF system deals with estimates from LTs in the same state space where the dependence between the estimation errors needs to be accounted for. Such a dependence arises from a common process noise which contributes to both errors since the common process model is shared by both trackers [1]. When the sensors are running asynchronously with different sampling intervals, the T2TF is carried out at the union of the sampling times (FC driven) and the cross-covariance for asynchronous and homogeneous T2TF was introduced in [4]. This present work considers a heterogeneous T2TF system with estimates from one radar and one infrared/electro-optical

(IR/EO) sensor. The radar provides estimates in Cartesian state space while the IR/EO sensor provides estimates in angular state space. Such a problem was first introduced in [3] where the assumption of estimation error independence was made as the cross-covariance between the estimation errors was not available — the cross-covariance was assumed to be zero. Ref. [5] deals with the heterogeneous T2TF from an IRST sensor and an air MTI radar but assuming the local state vectors have the same dimension with a unique inverse mapping, which is not realistic in most of the heterogeneous T2TF scenarios where the inverse transformation does not exist due to the state space dimension inequality.

The derivation of the cross-covariance for the heterogeneous LTs from one radar and one IR/EO sensor is provided. The two local states — one in Cartesian space with dimension 4 and one in angular space with dimension 2 — are related by a nonlinear relationship without inverse transformation. The relationship between the process noise covariances of the two motion models is presented. The cross-covariance of the estimation errors from two LTs is derived in a recursive form by taking into account the relationship between the local state model process noises. The synchronous T2TF fusion is carried out with a linear minimum mean square (LMMSE) estimator. The T2TF fusion with cross-covariance between the estimation errors is shown to be consistent. However, under the assumption of independence between the estimation errors, the T2TF is not consistent as the covariance is optimistic, which confirms the discussion in [1] for homogeneous case.

The paper is organized as follows. Section II formulates the state models and measurement models. In Section III, the linear minimum mean square error (LMMSE) estimator is introduced. Section IV derives the relationship of the process noise covariances as well as the cross-covariance of the estimation errors. Sec. V presents the simulation results from Monte-Carlo runs. The summary and conclusions are in Sec. VI.

II. THE STATE MODELS AND MEASUREMENT MODELS

The radar (located at $[\xi^r \ \eta^r]$) uses a Kalman Filter based on a nearly constant velocity (NCV) model (white noise

acceleration — WNA [2]) with a 4-d state vector comprising position and velocity

$$\mathbf{x}^r = [\xi \quad \dot{\xi} \quad \eta \quad \dot{\eta}] \quad (1)$$

The discretized dynamic model and the measurement model are

$$\mathbf{x}^r(k+1) = F^r \mathbf{x}^r(k) + \mathbf{v}^r(k) \quad (2)$$

$$\mathbf{z}^r(k) = H^r \mathbf{x}^r(k) + \mathbf{w}^r(k) \quad (3)$$

where

$$F^r = \begin{bmatrix} 1 & T^r & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T^r \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$H^r = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (5)$$

and T^r is the radar sampling interval.

The covariance of the discrete-time process noise is

$$Q^r = \begin{bmatrix} \frac{(T^r)^3}{3} & \frac{(T^r)^2}{2} & 0 & 0 \\ \frac{(T^r)^2}{2} & T^r & 0 & 0 \\ 0 & 0 & \frac{(T^r)^3}{3} & \frac{(T^r)^2}{2} \\ 0 & 0 & \frac{(T^r)^2}{2} & T^r \end{bmatrix} \tilde{q}^r \quad (6)$$

where $\tilde{q}^r$ is the process noise power spectral density.

The IR/EO sensor (located at $[\xi^e \quad \eta^e]$) uses a Kalman Filter based on a continuous time white noise acceleration (CWNA) model with a 2-d state vector comprising the angle and angle rate

$$\mathbf{x}^e = [\theta \quad \dot{\theta}] \quad (7)$$

The discretized dynamic model and the measurement model are

$$\mathbf{x}^e(k+1) = F^e \mathbf{x}^e(k) + \mathbf{v}^e(k) \quad (8)$$

$$\mathbf{z}^e = H^e \mathbf{x}^e(k) + \mathbf{w}^e(k) \quad (9)$$

where

$$F^e = \begin{bmatrix} 1 & T^e \\ 0 & 1 \end{bmatrix} \quad (10)$$

$$H^e = [1 \quad 0] \quad (11)$$

and T^e is the IR/EO sensor sampling interval.

The process noise covariance matrix of the IR/EO sensor's model at time k has the relationship with the radar process noise covariance matrix to be introduced in the Sec. IV. The state vector (1) and the state vector (7) have a nonlinear relationship, at the same sampling time (omitting the time argument, for simplicity),

$$\mathbf{x}^e = \alpha [\mathbf{x}^r] \quad (12)$$

with explicit expressions

$$\theta = \text{atan} \frac{\eta - \eta^e}{\xi - \xi^e} \quad (13)$$

$$\dot{\theta} = \frac{v \sin(\phi)}{r^e} \quad (14)$$

where v is the target speed

$$v = \sqrt{\dot{\xi}^2 + \dot{\eta}^2} \quad (15)$$

r^e is the range with respect to the passive sensor's location

$$r^e = \sqrt{(\xi - \xi^e)^2 + (\eta - \eta^e)^2} \quad (16)$$

and ϕ is the difference between velocity angle and position azimuth angle

$$\phi = \text{atan} \frac{\dot{\eta}}{\dot{\xi}} - \text{atan} \frac{\eta - \eta^e}{\xi - \xi^e} \quad (17)$$

In the ξ - η space, the radar has range and azimuth angle measurements (without time arguments, for simplicity)

$$r = \sqrt{(\xi - \xi^r)^2 + (\eta - \eta^r)^2} + w^r \quad (18)$$

$$\theta^r = \tan^{-1}[(\eta - \eta^r)/(\xi - \xi^r)] + w^\theta \quad (19)$$

and the IR/EO sensor has bearing (azimuth) measurements only

$$\theta^e = \tan^{-1}[(\eta - \eta^e)/(\xi - \xi^e)] + w^e \quad (20)$$

The measurement noises w^r , w^θ and w^e are assumed to be independent zero-mean white Gaussian with corresponding standard deviations σ^r , σ^θ and σ^e .

The radar's measurements in polar coordinates are transformed into Cartesian coordinates with an unbiased transformation [2]. Given (18) and (19), the unbiased transformed measurement vector is

$$\mathbf{z}_C = \begin{bmatrix} \xi_C \\ \eta_C \end{bmatrix} = \begin{bmatrix} b_1^{-1} r \cos(\theta^r) + \xi^r \\ b_1^{-1} r \sin(\theta^r) + \eta^r \end{bmatrix} \quad (21)$$

where

$$b_1 = e^{-\sigma^r/2} \quad (22)$$

The transformed radar noise vector $\mathbf{w}_C$ has the covariance matrix $\mathbf{R}_C$ with elements

$$\mathbf{R}_C(1,1) = b_1^{-2} r^2 \cos(\theta^r) + 0.5(r^2 + \sigma^r)(1 + b_1^4 \cos(\theta^r)) \quad (23)$$

$$\mathbf{R}_C(2,2) = b_1^{-2} r^2 \sin(\theta^r) + 0.5(r^2 + \sigma^r)(1 - b_1^4 \cos(\theta^r)) \quad (24)$$

$$\begin{aligned} \mathbf{R}_C(1,2) &= \mathbf{R}_C(2,1) \\ &= (0.5b_1^{-2} r^2 + 0.5(r^2 + \sigma^r)b_2 - r^2) \sin(2\theta^r) \end{aligned} \quad (25)$$

III. THE LMMSE ESTIMATOR FOR HETEROGENEOUS T2TF

The heterogeneous T2TF is carried out (omitting the time arguments) with a LMMSE estimator based on the estimate from radar

$$\hat{\mathbf{x}}^r = [\hat{\xi} \quad \hat{\dot{\xi}} \quad \hat{\eta} \quad \hat{\dot{\eta}}] \quad (26)$$

with corresponding covariance P^r and the estimate from IR/EO sensor

$$\hat{\mathbf{x}}^e = [\hat{\theta} \quad \hat{\dot{\theta}}] \quad (27)$$

with corresponding covariance P^e .

The fused track estimate in Cartesian coordinates is obtained, with details introduced in [3], as

$$\hat{\mathbf{x}}_{\text{LMMSE}}^r = \hat{\mathbf{x}}^r + P_{\mathbf{xz}} P_{\mathbf{zz}}^{-1} [\hat{\mathbf{x}}^e - g[\hat{\mathbf{x}}^r]] \quad (28)$$

where $g[\cdot]$ is the nonlinear relationship (12) between the states from the two sensors. Here we use g rather than α to avoid ambiguity in calculating the different Jacobians needed in the sequel. The corresponding fused covariance matrix is

$$P_{\text{LMMSE}}^r = P^r - P_{\mathbf{xz}} P_{\mathbf{zz}}^{-1} P_{\mathbf{xz}}' \quad (29)$$

where

$$P_{\mathbf{xz}} \approx P^r (G^r)' - P^{re} \quad (30)$$

$$P_{\mathbf{zz}} \approx P^r - G^r P^{re} - P^{er} (G^r)' + G^r P^r (G^r)' \quad (31)$$

and

$$G^r = [\nabla_{\mathbf{x}} g(\mathbf{x}^e)]' |_{\mathbf{x}^r = \hat{\mathbf{x}}^r} \quad (32)$$

is the Jacobian evaluated at the estimate from the radar. It can be seen from the above equations, the key to the T2TF problem is to find the cross-covariance between the estimation errors P^{re} , which is derived in the following section.

IV. CROSS-COVARIANCE BETWEEN ESTIMATION ERRORS

A. Relationship Between the Process Noises

When the sensors are running synchronously, the LTs share the same sampling time. The LTs have full rate communication with the Fusion Center (FC). For the heterogeneous state models and measurement models introduced in Sec. II, the process noises and measurement noises assumed to be additive, zero mean and white with corresponding covariance matrices $Q^m(k)$ and $R^m(k)$ ($m = r, e$). All the noises are also assumed to be mutually independent except $\mathbf{v}^r$ is correlated with $\mathbf{v}^e$ since they pertain to the motion of the same target, although in different state spaces. The recursion of the cross-covariance $Q^{re}(k)$ between $\mathbf{v}^r(k)$ and $\mathbf{v}^e(k)$ will be discussed later.

The two state vectors have a nonlinear relationship as shown in (12), which has no inverse transformation due to the dimension inequality.

Substituting (2) into the (12) yields

$$\mathbf{x}^e(k+1) = \alpha[\mathbf{x}^r(k+1)] = \alpha[F^r \mathbf{x}^r(k) + \mathbf{v}^r(k)] \quad (33)$$

The vector Taylor series expansion of (33) up to the first-order term is

$$\alpha[\mathbf{x}^r(k+1)] = \alpha[F^r \mathbf{x}^r(k)] + [\nabla_x \alpha(x)]' |_{x=F^r \mathbf{x}^r(k)} \mathbf{v}^r(k) \quad (34)$$

Thus

$$\mathbf{v}^e(k) \approx [\nabla_x \alpha(x)]' |_{x=F^r \mathbf{x}^r(k)} \mathbf{v}^r(k) = A(k) \mathbf{v}^r(k) \quad (35)$$

where

$$A(k) \triangleq [\nabla_x \alpha(x)]' |_{x=F^r \mathbf{x}^r(k)} \quad (36)$$

is the Jacobian corresponding to (35). The elements in (36) are

$$A(1,1) = -\frac{\eta - \eta^e}{(r^e)^2} \quad (37)$$

$$A(1,2) = 0 \quad (38)$$

$$A(1,3) = -\frac{\xi - \xi^e}{(r^e)^2} \quad (39)$$

$$A(1,4) = 0 \quad (40)$$

$$A(2,1) = \frac{v}{(r^e)^3} [-\sin(\phi)(\xi - \xi^e) + \cos(\phi)(\eta - \eta^e)] \quad (41)$$

$$A(2,2) = \frac{1}{vr^e} [\sin(\phi)\dot{\xi} - \cos(\phi)\dot{\eta}] \quad (42)$$

$$A(2,3) = \frac{v}{(r^e)^3} [-\sin(\phi)(\xi - \xi^e) - \cos(\phi)(\eta - \eta^e)] \quad (43)$$

$$A(2,4) = \frac{1}{vr^e} [\sin(\phi)\dot{\eta} + \cos(\phi)\dot{\xi}] \quad (44)$$

where v is defined in (15), r^e is defined in (16) and ϕ is defined in (17).

Then, $Q^e(k)$, the covariance of the process noise $\mathbf{v}^e(k)$ can be expressed using $Q^r(k)$ as

$$Q^e(k) = E[\mathbf{v}^e(k) \mathbf{v}^e(k)'] = A(k) Q^r(k) A(k)' \quad (45)$$

and the cross-covariance between the two process noises is

$$Q^{re}(k) = E[\mathbf{v}^r(k) \mathbf{v}^e(k)'] = Q^r(k) A(k)' \quad (46)$$

Note that in the estimation problem where the state is not available, the Jacobian (36) will have to be evaluated at the latest estimate.

B. Relationship Between the Local Estimation Errors

The updated states for both radar and IR/EO sensor at time k are, using a Kalman Filter,

$$\hat{\mathbf{x}}^r(k|k) = F^r \hat{\mathbf{x}}^r(k-1|k-1) + W^r(k) \nu_r(k) \quad (47)$$

$$\hat{\mathbf{x}}^e(k|k) = F^e \hat{\mathbf{x}}^e(k-1|k-1) + W^e(k) \nu_e(k) \quad (48)$$

where the innovations are

$$\begin{aligned} \nu^r(k) &= H^r(k) [\mathbf{x}^r(k) - \hat{\mathbf{x}}^r(k|k-1)] + \mathbf{w}^r(k) \\ &= H(k)^r F^r(k-1) \tilde{\mathbf{x}}^r(k-1|k-1) \\ &\quad + H^r(k) \mathbf{v}^r(k-1) + \mathbf{w}^r(k) \end{aligned} \quad (49)$$

and

$$\begin{aligned} \nu^e(k) &= H^e(k) [\mathbf{x}^e(k) - \hat{\mathbf{x}}^e(k|k-1)] + \mathbf{w}^e(k) \\ &= H(k)^e F^e(k-1) \tilde{\mathbf{x}}^e(k-1|k-1) \\ &\quad + H^e(k) \mathbf{v}^e(k-1) + \mathbf{w}^e(k) \end{aligned} \quad (50)$$

where the second lines of the above equations are obtained by substituting the dynamic equations (1) and (8) into the first lines, respectively.

The estimation errors from radar and EO/IR sensor at time k can be expressed as

$$\begin{aligned} \tilde{\mathbf{x}}^r(k|k) &= \mathbf{x}^r(k) - \hat{\mathbf{x}}^r(k|k) \\ &= [I_{n^r} - W^r(k) H^r(k)] F^r(k-1) \tilde{\mathbf{x}}^r(k-1|k-1) \\ &\quad + [I_{n^r} - W^r(k) H^r(k)] \mathbf{v}^r(k-1) + W^r(k) \mathbf{w}^r(k) \end{aligned} \quad (51)$$

and

$$\begin{aligned}\tilde{\mathbf{x}}^e(k|k) &= \mathbf{x}^e(k) - \hat{\mathbf{x}}^e(k|k) \\ &= [I_{n^e} - W^e(k)H^e(k)]F^e(k-1)\tilde{\mathbf{x}}^e(k-1|k-1) \\ &\quad + [I_{n^e} - W^e(k)H^e(k)]\mathbf{v}(k-1) + W(k)^e\mathbf{w}^e(k)\end{aligned}\quad (52)$$

where I_{n^r} is the n^r -dimensional identity matrix and n^r is the dimension of the state vector $\mathbf{x}^r$. Similarly for I_{n^e} .

Then the estimation errors' cross-covariance (of dimension $n^r \times n^e$) is

$$\begin{aligned}P^{\text{re}}(k|k) &= E[\tilde{\mathbf{x}}^r(k|k)\tilde{\mathbf{x}}^e(k|k)'] \\ &= [I - W^r(k)H^r(k)]F^r(k-1)P^{\text{re}}(k-1|k-1)F^e(k-1)' \\ &\quad \cdot [I - W^e(k)H^e(k)]' + [I - W^r(k)H^r(k)]Q^{\text{re}}(k-1) \\ &\quad \cdot [I - W^e(k)H^e(k)]' \\ &= [I - W^r(k)H^r(k)]\{F^r(k-1)P^{\text{re}}(k-1|k-1)F^e(k-1)' \\ &\quad + Q^{\text{re}}(k-1)\}[I - W^e(k)H^e(k)]'\end{aligned}\quad (53)$$

With (53), the T2TF can be carried out using the LMMSE introduced in Sec. III.

C. Fusion Center Driven T2TF

For FC driven T2TF, the fusion is carried out at certain frequency different from the local sensors, which are running synchronously, i.e., with same sampling interval. Denote by n the time index of the FC and the cross-covariance between the process noises (using two arguments to denote the sampling interval at the FC) is

$$Q^{\text{re}}[t_n, t_{n-1}] = Q^r[t_n, t_{n-1}] A[t_n, t_{n-1}, \mathbf{x}^e(t_{n-1})] \quad (54)$$

where

$$A[t_n, t_{n-1}, \mathbf{x}^e(t_{n-1})] \triangleq [\nabla_x \alpha(x)']|_{x=F^r[t_n, t_{n-1}]\mathbf{x}^r(t_{n-1})} \quad (55)$$

The cross-covariance of the estimation errors, obtained by modifying (53), can be written as

$$\begin{aligned}P^{\text{re}}(t_n|t_n) &= [I - W^r(t_n)H^r(t_n)] \\ &\quad \cdot \{F^r[t_n, t_{n-1}]P^{\text{re}}(t_{n-1}|t_{n-1})F^e[t_n, t_{n-1}]' + Q^{\text{re}}[t_n, t_{n-1}]\} \\ &\quad \cdot [I - W^e(t_n)H^e(t_n)]'\end{aligned}\quad (56)$$

V. SIMULATION RESULTS

In the simulation, a target moving on ξ - η plane is considered with initial position $[-5 \ 10]$ km and the initial velocity $[20 \ 200]$ m/s in Cartesian coordinates. The process noise induced root mean square (RMS) acceleration over a sampling interval T is

$$d \triangleq \frac{\sqrt{qT}}{T} \quad (57)$$

with value $d = 5 \text{ m/s}^2$. The active sensor is located at $[6.9 \ 5.4]$ km with measurement noise standard deviations $\sigma^r = 20 \text{ m}$ and $\sigma^\theta = 1 \text{ mrad}$. The passive sensor is located at $[3.8 \ 1.9]$ km with angle measurement noise $\sigma^e = 1 \text{ mrad}$. The

sampling intervals are $T^r = T^e = 1 \text{ s}$ and the trajectory lasts for 100 s. The target trajectory and the sensor positions are shown in Fig. 1. The following simulation results are obtained from 500 Monte-Carlo (MC) runs. The fusion is carried out after the convergence of the local trackers.

Fig. 1: Target trajectory on the ξ - η plane

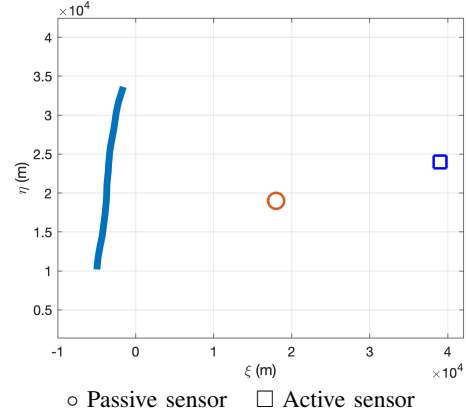


Fig. 2: Position RMSE from 500 MC runs

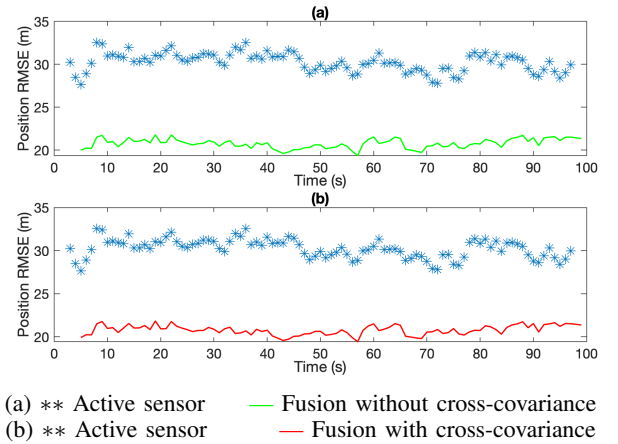


Fig. 2 shows the root mean square errors (RMSE) of the position vector from both the active sensor, fusion without cross-covariance and fusion with cross-covariance. Similarly, the velocity vector RMSEs are shown in Fig. 3. It can be seen that the cross-covariance does not have a significant impact on the RMSE for both the position and velocity. However, consistency is another crucial standard to evaluate a system, which is shown in the sequel.

For the synchronous, heterogeneous case considered, the fusion with cross-covariance yields the actual variance. The variance differences (variance of fusion with cross-covariance minus variance of fusion without cross-covariance) for each component in the Cartesian state vector are shown in Fig. 4. The consistency is shown in Fig. 5 through the normalized estimation error squared (NEES) for both the fusion with cross-covariance and the fusion without cross-covariance. Given a

Fig. 3: Velocity RMSE from 500 MC runs

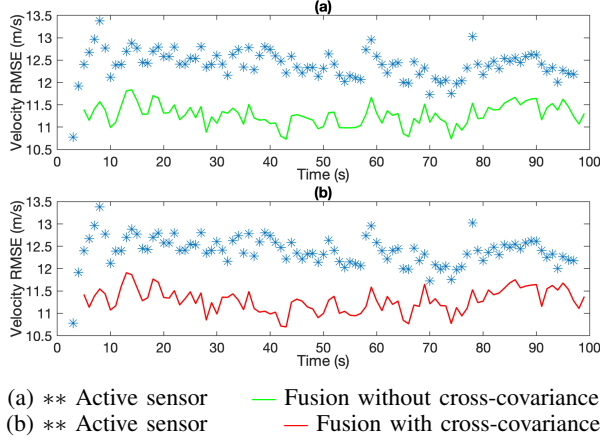
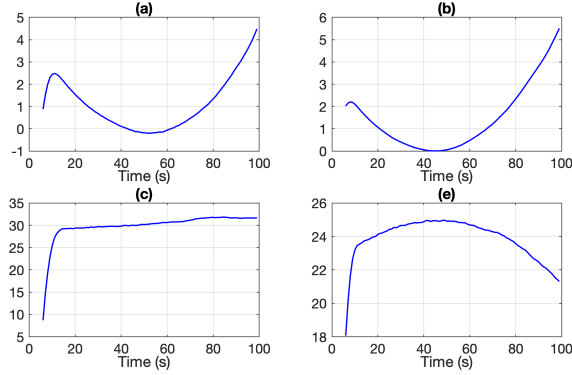


Fig. 4: Variance Difference of elements from the fused state estimate in Cartesian Coordinates (with cross-covariance minus without cross-covariance)



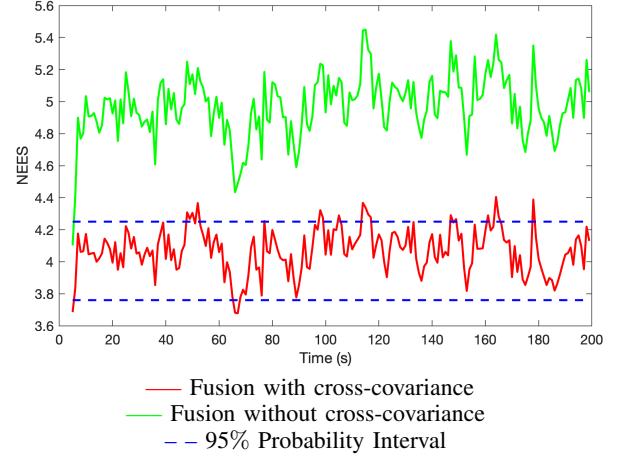
(a) Variance Difference of ξ (m)² (b) Variance Difference of $\dot{\xi}$ (m/s)²
(c) Variance Difference of η (m)² (d) Variance Difference of $\dot{\eta}$ (m/s)²

95% confidence level, the probability interval for such a 4-dimensional state vector from 500 Monte-Carlo runs is [3.76 4.25]. The system is consistent if the NEES falls into the above interval. It can be seen that neglecting the cross-covariance between the estimation errors makes the fusion inconsistent and the covariance is optimistic.

VI. SUMMARY AND CONCLUSIONS

In this work, we derived the cross-covariance between the local estimation errors of heterogeneous tracks from one radar and one IR/EO sensor with different models. The radar provides an estimate in Cartesian state space while with IR/EO sensor provides an estimate in angular state space. The simulation results show the performance of both the heterogeneous track-to-track fusion with cross-covariance and without cross-covariance. It can be seen that with the cross-covariance, the T2TF can achieve improved performance by making the system consistent. Without the cross-covariance involved, although the RMSE has minor difference, the fused

Fig. 5: NEES from 500 MC runs



covariance is not accurate which makes the heterogeneous T2TF inconsistent.

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